Zecpath AI System Architecture

High-Level Architecture

Frontend

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Backend APIs

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AI Services

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Database

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Storage

AI Services

- Resume Parser
- ATS Engine
- Screening AI
- HR AI
- Technical AI
- Machine Test AI
- Behavioral AI
- Integrity AI
- Aggregation Engine
- Decision Engine
- Report Generator

Data Flow

Candidate Upload

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Resume Parsing

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ATS Matching

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Screening

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HR Interview

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Technical Interview

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Machine Test

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Behavior Analysis

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Aggregation

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Decision

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Hiring Report

Component Responsibilities

Resume Parser

Extracts structured information from resumes.

ATS Engine

Calculates resume-job matching score.

Screening AI

Evaluates candidate responses.

HR AI

Measures communication and HR competencies.

Technical AI

Evaluates technical knowledge.

Machine Test AI

Evaluates coding and practical tasks.

Behavioral AI

Detects engagement and behavioral signals.

Integrity AI

Detects cheating or suspicious behavior.

Aggregation Engine

Combines all stage scores.

Decision AI

Generates final hiring recommendation.

Report Generator

Creates recruiter-friendly hiring reports.

Database Objects

- Candidate
- Resume
- Job Description
- Interview Session
- Scores
- Reports
- Audit Logs

API Flow

Frontend

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Backend

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AI APIs

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Database

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Frontend Response

Deployment Architecture

Client

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NGINX

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Backend APIs

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AI Services

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Redis Cache

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PostgreSQL

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Monitoring

Scalability

- Microservices
- Horizontal Scaling
- Load Balancer
- Docker Containers
- Kubernetes

Monitoring

- Logging
- Metrics
- Alerts
- Tracing
- Audit Logs